

RSS and Model Comparison

Sum of squared residuals, nested models, sequential ANOVA

STAT 24510-30040

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Sum of squared residuals of a linear model, denoted as RSS or RSS_{Model} or MSS , is a measure on how good the proposed model explains variations in the observed data, that is, it's a measure of the goodness-of-fit of the model.

RSS is widely used to compare the relative goodness-of-fit of nested models.

For example, analysis of variance is carried out by comparing RSS of different models.

Definition

- Fitted value — Estimate of the mean of an observation, e.g., the fitted value $\hat{y}_i$ is the model estimated mean of observation y_i .
- Residual — Difference between observation and fitted value, e.g. $(y_i - \hat{y}_i)$

Naming conventions

- Residual sum of squares, RSS, e.g. $\sum_i (y_i - \hat{y}_i)^2$, where the sum is over all observations.
- Other names: Sum of squared errors (SSE, SS_{error}), Sum of squared residuals (SSR, $SS_{residuals}$).

In the following, we investigate RSS properties by examples, and demonstrate the use of RSS in nested models, especially, in sequential analysis of variance (sequential ANOVA) for unbalanced data.

A derivation of the probability distribution of RSS under normality assumption is in the appendix.

RSS example — Two independent samples

Data: Two sets of observations or measurements: $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_m\}$.

Model assumptions: Independent sample with equal variance.

- $x_i, \dots, x_n$ are i.i.d. observations from a distribution with mean μ_x , variance σ^2 .
- $y_1, \dots, y_m$ are i.i.d. observations from a distribution with mean μ_y , variance σ^2 .
- The two samples $\{x_i, \dots, x_n\}$ and $\{y_1, \dots, y_m\}$ are independent.

The standard two-sample procedure is equivalent to testing the hypotheses

$$\begin{cases} H_o : \mu_x = \mu_y = \mu_o \\ H_1 : \mu_x \neq \mu_y \end{cases}$$

Each hypothesis represents a model, with its parameter estimates given data.

Under H_o , every observation has fitted value $\hat{\mu}_o$, the estimated mean of any and all observations, x_i or y_i .

$$H_o : \mu_x = \mu_y = \mu_o \quad \text{corresponding to the null model} \quad \begin{cases} X_i = \mu_o + \varepsilon_i, & i = 1, \dots, n, \\ Y_j = \mu_o + \varepsilon_j, & j = 1, \dots, m. \end{cases}$$

Therefore the model under the null hypothesis H_o has residual sum of squares

$$RSS_o = \sum_{i=1}^n (x_i - \hat{\mu}_o)^2 + \sum_{j=1}^m (y_j - \hat{\mu}_o)^2$$

where $\hat{\mu}_o$ is the model estimate of μ_o , also called the fitted value by the null model.

Under $H_1 \cup H_o$, the union of the alternative hypothesis H_1 and the null H_o , the corresponding linear model is

$$\begin{cases} X_i = \mu_1 + \varepsilon_i, & i = 1, \dots, n, \\ Y_j = \mu_2 + \varepsilon_j, & j = 1, \dots, m. \end{cases}$$

Under $H_1 \cup H_o$, the model fitted value for observation x_i is $\hat{\mu}_x$, while y_i has a different fitted value $\hat{\mu}_y$.

The residual sum of squares for the model under $H_1 \cup H_o$ is

$$RSS_1 = \sum_{i=1}^n (x_i - \hat{\mu}_x)^2 + \sum_{j=1}^m (y_j - \hat{\mu}_y)^2$$

To use distribution based method such as the likelihood ratio test, we further assume normal distribution,

- $x_i, \dots, x_n$ are i.i.d. observations from $N(\mu_1, \sigma^2)$,
- $y_1, \dots, y_m$ are i.i.d. observations from $N(\mu_2, \sigma^2)$,
- The two samples $\{x_i, \dots, x_n\}$ and $\{y_1, \dots, y_m\}$ are independent.

As we have derived previously, the maximum likelihood estimates (MLE) for the mean parameters are the same as the estimates by the method of least squares (LSE).

The MLE for the mean parameter under $H_o : \mu_X = \mu_Y = \mu_o$ is

$$\hat{\mu}_o = \frac{\sum_{i=1}^n x_i + \sum_{i=1}^m y_i}{n + m} = \frac{n\bar{x} + m\bar{y}}{n + m}$$

MLEs for the mean parameters for $H_1 \cup H_o$ are

$$\hat{\mu}_1 = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}, \quad \hat{\mu}_2 = \frac{\sum_{i=1}^m y_i}{m} = \bar{y}$$

The residual sum of squares for the model under H_o is the SS_E under the null model,

$$RSS_o = \sum_{i=1}^n (x_i - \hat{\mu}_o)^2 + \sum_{i=1}^m (y_i - \hat{\mu}_o)^2$$

The residual sum of squares for the model under $H_a \cup H_o$ is

$$RSS_1 = \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^m (y_i - \bar{y})^2$$

To see the relation between the residual sums of squares RSS_1 and RSS_o of the two models, decompose the first term in RSS_o by the sample mean,

$$\begin{aligned}\sum_{i=1}^n (x_i - \hat{\mu}_o)^2 &= \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - \hat{\mu}_o)^2 \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^n (\bar{x} - \hat{\mu}_o)^2 + 2 \sum_{i=1}^n (x_i - \bar{x})(\bar{x} - \hat{\mu}_o) \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \hat{\mu}_o)^2\end{aligned}$$

Similarly, the second term in RSS_o can be written as

$$\sum_{i=1}^m (y_i - \hat{\mu}_o)^2 = \sum_{i=1}^m (y_i - \bar{y})^2 + m(\bar{y} - \hat{\mu}_o)^2$$

Therefore,

$$\begin{aligned}RSS_o &= \sum_{i=1}^n (x_i - \hat{\mu}_o)^2 + \sum_{i=1}^m (y_i - \hat{\mu}_o)^2 \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^m (y_i - \bar{y})^2 + n(\bar{x} - \hat{\mu}_o)^2 + m(\bar{y} - \hat{\mu}_o)^2 \\ &= RSS_1 + n(\bar{x} - \hat{\mu}_o)^2 + m(\bar{y} - \hat{\mu}_o)^2 \\ &= RSS_1 + \sum_{i=1}^n (\bar{x} - \hat{\mu}_o)^2 + \sum_{i=1}^m (\bar{y} - \hat{\mu}_o)^2 \\ &= RSS_1 + \sum_{\text{Observations}} (\text{fitted-value}_1 - \text{fitted-value}_o)^2\end{aligned}$$

where the sum is over all observations. For each observation,

- The “fitted-value₁” is the model estimated value for the observation by the model under the joint $H_1 \cup H_0$, corresponding to the relatively more complicated model,
- The “fitted-value₀” is the model estimated value for the observation by the model under the null hypothesis H_0 , corresponding to the simpler model, which is a special case, a sub-model of the model under $H_1 \cup H_0$.

There is always the relation

$$RSS_1 \leq RSS_o$$

That is, the RSS under the simpler, smaller model is always large than or equal to the RSS under the more complicated, larger model, which includes the smaller model as a special case.

The added complexity moves the larger model closer to the observed data.

RSS and LR

Recall in likelihood ratio test (LRT) for two independent normal samples with equal variance, the ratio of the maximum likelihood is

$$LR = \frac{\max_{H_o} \ell}{\max_{H_a \cup H_o} \ell} = \left(\frac{\hat{\sigma}_a^2}{\hat{\sigma}_o^2} \right)^{(n+m)/2}$$

The likelihood ratio test statistic

$$T = -2 \log(LR) = (n+m) \log \frac{\hat{\sigma}_o^2}{\hat{\sigma}_a^2} \sim \chi_1^2 \quad \text{under } H_o$$

asymptotically, when the sample sizes n and m both are large.

$$\text{Reject the null } H_o \text{ when } T \text{ is large} \iff \text{Reject the null when } \frac{\hat{\sigma}_o^2}{\hat{\sigma}_a^2} \text{ is large}$$

Recall the MLE for the variance under H_o is

$$\hat{\sigma}_o^2 = \frac{\sum_{i=1}^n (x_i - \hat{\mu}_o)^2 + \sum_{i=1}^m (y_i - \hat{\mu}_o)^2}{n+m} = \frac{RSS_o}{n+m}$$

The MLE for the variance under $H_1 \cup H_0$ is

$$\hat{\sigma}_a^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^m (y_i - \bar{y})^2}{n+m} = \frac{RSS_1}{n+m}$$

Hence the LR test statistic is also based on comparison of the RSS since

$$\frac{\hat{\sigma}_o^2}{\hat{\sigma}_a^2} = \frac{RSS_o}{RSS_1}$$

In the context of linear models, the two-sample test corresponds to the model

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, 2; \quad j = 1, \dots, n.$$

The hypotheses tested above are equivalent to

$$\begin{cases} H_o : & \alpha_1 = \alpha_2 \\ H_1 : & \alpha_1 \neq \alpha_2 \end{cases}$$

The model is over-parameterized, as we discussed earlier. However the hypotheses stated stay the same (i.e. equivalent) under any common parameter constraints.

RSS example — One-way layout analysis of variance

Data The observations are g sets of measurements.

$$\begin{aligned}&\{y_{11}, \dots, y_{1,J_1}\} \\ &\{y_{21}, \dots, \dots, \dots, y_{2,J_2}\} \\ &\vdots \\ &\{y_{g1}, \dots, \dots, y_{g,J_g}\}\end{aligned}$$

Assumptions

- Measurements $y_{ij}, j = 1, \dots, J_i$ are independent observations from a distribution with mean μ_i , variance σ^2 .
- The g samples are independent of each other.

Hypothesis testing related to the one-way analysis of variance is

$$\begin{cases} H_o : & \mu_1 = \mu_2 = \cdots = \mu_g = \mu \\ H_a : & \mu_i \neq \mu_j \quad \text{for some } i, j = 1, \dots, g. \end{cases}$$

With added normality assumption, the ANOVA linear model is

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim i.i.d. N(0, \sigma^2)$$

The linear model can be re-parametrized as

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim i.i.d. N(0, \sigma^2)$$

The equivalent hypothesis test is

$$\begin{cases} H_o : & \alpha_1 = \alpha_2 = \cdots = \alpha_g \\ H_a : & \alpha_i \neq \alpha_j \quad \text{for some } i, j = 1, \dots, g. \end{cases}$$

The model written in terms of relative or differential effect α_i 's is over-parameterized, so some constraints are needed to make the parameters identifiable. Albeit the hypotheses are the same under any common parameter constraints.

Under the normality assumption and proper constraints, MLEs can be obtained for parameters in the linear model.

Analogous to the two-sample cases, the MLE for the variance under the null H_o is

$$\hat{\sigma}_o^2 = \frac{1}{n} \sum_{i=1}^g \sum_{j=1}^{J_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2 = \frac{1}{n} RSS_o$$

The variance MLE under the alternative H_a is

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^g \sum_{j=1}^{J_i} (y_{ij} - \bar{y}_{i\bullet})^2 = \frac{1}{n} RSS_1$$

Partition the squares,

$$\begin{aligned} RSS_o &= \sum_{i=1}^g \sum_{j=1}^{J_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2 = \sum_{i=1}^g \sum_{j=1}^{J_i} (y_{ij} - \bar{y}_{i\bullet} + \bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 \\ &= \sum_{i=1}^g \sum_{j=1}^{J_i} (y_{ij} - \bar{y}_{i\bullet})^2 + \sum_{i=1}^g \sum_{j=1}^{J_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 + 2 \sum_{i=1}^g \left(\sum_{j=1}^{J_i} (y_{ij} - \bar{y}_{i\bullet}) \right) (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet}) \\ &= \sum_{i=1}^g \sum_{j=1}^{J_i} (y_{ij} - \bar{y}_{i\bullet})^2 + \sum_{i=1}^g \sum_{j=1}^{J_i} (\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2 \end{aligned}$$

where we used $\sum_{j=1}^{J_i} (y_{ij} - \bar{y}_{i\bullet}) = 0$. This is the familiar decomposition in the context of ANOVA,

$$SS_{total} = SS_{error} + SS_{treatment}$$

Now in terms of sums of squared residuals RSS of different models, we have obtained a similar relationship between RSS_1 and RSS_o for ANOVA models.

$$\begin{aligned} RSS_o &= RSS_1 + SS_{treatment} \\ &= RSS_1 + \sum_{\text{observations}} (\text{fitted-value}_1 - \text{fitted-value}_o)^2 \end{aligned}$$

Just as in the two-sample case, the RSS under the smaller model is greater than or equal to the RSS under the larger model containing the smaller model,

$$RSS_o \geq RSS_1$$

Remarks: From vector-matrix projection point of view in matrix algebra, in $\mathbb{R}^n$, the residual vector is orthogonal to the vector of changes in fitted values, and both RSS_o and RSS_1 (residual) vectors are orthogonal to the vector of fitted values under the null model. We will learn the projection concept in more detail when we study the theory of linear models.

RSS example — Two-way layout analysis of variance

The data are y_{ijk} , where $i = 1, \dots, I$ are levels of treatment factor A, $j = 1, \dots, J$ are levels of treatment factor B, and $k = 1, \dots, K_{ij}$ are replicates of treatment (i, j) , the combination of level i of treatment factor A and level j of treatment factor B.

The null model

$$Y_{ijk} = \mu + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

is the model corresponding to

$$H_o : \mu_{ij} \equiv \mu$$

The MLE or LSE for the mean parameter is

$$\hat{\mu}_{ij} \equiv \hat{\mu} = \bar{y}_{\bullet\bullet\bullet}$$

The sum of squared residuals for the null model is

$$RSS_o = \sum_{i,j,k} (y_{ijk} - \hat{\mu})^2 = \sum_{i,j,k} (y_{ijk} - \bar{y}_{\bullet\bullet\bullet})^2$$

There are many options of alternative hypotheses. If the alternative hypothesis is

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$$

The corresponding linear model includes main effects and interaction,

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

The MLE for cell mean parameters is

$$\hat{\mu}_{ij} = \bar{y}_{ij\bullet}$$

The model residual sum of squares is

$$RSS_1 = \sum_{i,j,k} (y_{ijk} - \hat{\mu}_{ij})^2 = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\bullet})^2$$

Compared with the null model

$$Y_{ijk} = \mu + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

The null model's residual sum of squares can be written as

$$\begin{aligned} RSS_o &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{\bullet\bullet\bullet})^2 = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\bullet} + \bar{y}_{ij\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 \\ &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\bullet})^2 + \sum_{i,j,k} (\bar{y}_{ij\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 + 2 \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\bullet}) (\bar{y}_{ij\bullet} - \bar{y}_{\bullet\bullet\bullet}) \\ &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\bullet})^2 + \sum_{i,j,k} (\bar{y}_{ij\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 \\ &= RSS_1 + \sum_{\text{obs}} (\text{fitted-value}_1 - \text{fitted-value}_o)^2 \end{aligned}$$

where we applied $\sum_k (y_{ijk} - \bar{y}_{ij\bullet}) = 0$.

If the alternative hypothesis with respect to the null $H_0 : \mu_{ij} \equiv \mu$ is

$$\mu_{ij} = \mu + \alpha_i$$

The corresponding linear model has the main effect of factor A only,

$$Y_{ijk} = \mu + \alpha_i + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

The common estimator for cell means is

$$\hat{\mu}_{ij} = \bar{y}_{i\bullet\bullet}$$

The sum of squared residuals of the model is

$$RSS_1 = \sum_{i,j,k} (y_{ijk} - \bar{y}_{i\bullet\bullet})^2 = SS_A$$

Decompose the null model's residual sum of squares,

$$\begin{aligned} RSS_0 &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{\bullet\bullet\bullet})^2 \\ &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{i\bullet\bullet} + \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 \\ &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{i\bullet\bullet})^2 + \sum_{i,j,k} (\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 + 2 \sum_{i,j,k} (y_{ijk} - \bar{y}_{i\bullet\bullet}) (\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet}) \\ &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{i\bullet\bullet})^2 + \sum_{i,j,k} (\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 \\ &= RSS_1 + \sum_{\text{obs}} (\text{fitted-value}_1 - \text{fitted-value}_0)^2 \end{aligned}$$

where the sum of cross product term = 0 because, by definition of the cell mean, $\sum_{j,k} (y_{ijk} - \bar{y}_{ij\bullet}) = 0$.

RSS for balanced design

For balanced design with K replicates for each treatment combination,

$$\sum_{i,j,k} (y_{ijk} - \bar{y}_{\bullet\bullet\bullet})^2 = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\bullet})^2 + SS_A + SS_B + SS_{AB} \quad (1)$$

with

$$\begin{aligned} SS_A &= \sum_{i,j,k} (\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 = KJ \sum_i (\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 \\ SS_B &= \sum_{i,j,k} (\bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 = KI \sum_j (\bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 \\ SS_{A*B} &= \sum_{i,j,k} (\bar{y}_{ij\bullet} - \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet j\bullet} + \bar{y}_{\bullet\bullet\bullet})^2 = K \sum_{i,j} (\bar{y}_{ij\bullet} - \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet j\bullet} + \bar{y}_{\bullet\bullet\bullet})^2 \end{aligned}$$

The additive relationship is reflected in the balanced two-way ANOVA table.

Source	d.f.	SS	MS	Variance ratio (F)
A (I-levels)	$I - 1$	SS_A	$SS_A/(I - 1)$	MS_A/MS_E
B (J-levels)	$J - 1$	SS_B	$SS_B/(J - 1)$	MS_B/MS_E
A*B	$(I - 1)(J - 1)$	SS_{AB}	$SS_{AB}/[(I - 1)(J - 1)]$	MS_{AB}/MS_E
Error	$(K - 1)IJ$	SS_E	$SS_E/[(K - 1)IJ]$	
Total	$IJK - 1$	SS_{Total}		

Note that, for balanced design with equal number of replicates for every treatment combination, the order of Factor A and Factor B in the analysis of variance does not matter. The ANOVA table can be written as

Source	d.f.	SS	MS	Variance ratio (F)
B (J-levels)	$J - 1$	SS_B	$SS_B/(J - 1)$	MS_B/MS_E
A (I-levels)	$I - 1$	SS_A	$SS_A/(I - 1)$	MS_A/MS_E
A*B	$(I - 1)(J - 1)$	SS_{AB}	$SS_{AB}/[(I - 1)(J - 1)]$	MS_{AB}/MS_E
Error	$(K - 1)IJ$	SS_E	$SS_E/[(K - 1)IJ]$	
Total	$IJK - 1$	SS_{Total}		

The two ANOVA tables contain equivalent information, corresponding entries, such as SS_A have the same value.

Unbalanced two-way layout and sequential ANOVA

When the replicates for different treatment combinations are not equal, that is, when $K_{ij} \neq K$, the design is **unbalanced**.

The SS decomposition in (1) no longer holds for unbalanced data.

A common approach is to adjust the sum of squares so that all sums of squares still add up to the total sum of squares in the ANOVA table. This is the method used in **sequential** analysis of variance.

The consequence is that now the order of the factors in the ANOVA matters.

The SS values are not the same in ANOVA table with factor A first B second and the ANOVA with B first A second.

Example of unbalanced design (Sprouting barley) — Unbalanced two-factor model

Data: The growth of barley under the effects of water level and timing of planting the seedlings in the field.

The response is "seed", the two factors are "water" and "week".

The data table below show numbers of sprouts under each treatment combination, with three or two replicates.

water	week1	week3	week6	week9	week12
4	11	7	9	13	20
	9	16	19	35	37
	6	17	35	28	
8	8	1	5	1	11
	3	7	9	10	15
	3	3	9	9	25

Note that in "water-4" and "week-12" combination, there are only two replicates, while there are three replicates in all other treatment combinations.

The unbalanced data has the following table of averages:

water	Means of seed					water mean
	1	3	6	9	12	
4	8.7	13.3	21.0	25.3	28.5	18.7
8	4.7	3.7	7.7	6.7	17.0	7.93
week mean	6.7	8.5	14.3	16.0	21.6	13.1

Consider the two-factor linear models

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

and

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

where i denotes the two levels “4” and “8” of the factor “water”, j indexes the five levels of the factor “week”, and $k = 1, 2, 3$ labels the replicates within each treatment combination of “water” level i and “week” level j .

The independence assumption is likely true for this example, the normality assumption is an approximation.

In the following we investigate analysis of variance on the data.

- Enter factor “water” first in analysis of variance

Consider entering the factor “week” first then factor “water” in ANOVA.

```
aov(seed ~ week + water)
aov(seed ~ week* water)
```

Below are ANOVA tables corresponding to the two-factor linear model with the interaction term and the one without the interaction term.

Source	(seq)SS	df	MS	V-ratio(F)	Prob> F
week	796.1	4	199.0	3.732	0.0210
water	940.7	1	940.7	17.642	0.0005
week*water	171.5	4	42.9	0.804	0.5378
Residual	1013.1	19	53.3		
Total	2921.5	28	104.3		

Source	(seq)SS	df	MS	V-ratio(F)	Prob> F
week	796.1	4	199.0	3.864	0.0152
water	940.7	1	940.7	18.265	0.0003
Residual	1184.6	23	51.5		
Total	2921.5	28	104.3		

- Enter factor “water” first in analysis of variance

When the variable order is factor “water” followed by factor “week”, the ANOVA tables for the models with interaction and without interaction are below.

Source	(seq)SS	df	MS	V-ratio(F)	Prob> F
water	841.7	1	841.7	15.784	0.0008
week	895.2	4	223.8	4.197	0.0133
week*water	171.5	4	42.9	0.804	0.5378
Residual	1013.2	19	53.3		
Total	2921.5	28	104.3		

Source	(seq)SS	df	MS	V-ratio(F)	Prob> F
water	841.7	1	841.7	15.784	0.0005
week	895.2	4	223.8	4.345	0.0092
Residual	1184.6	23	51.5		
Total	2921.5	28	104.3		

- Comparison with induced one-way analysis of variance

For comparison, we show the induced one-way analysis of variance for each of the two factors, and examine their sum of squares.

Source	(seq)SS	df	MS	V-ratio(F)	Prob> F
week	796.1	4	199.02	2.247	0.0939
Residual	2125.4	24	88.56		
Total	2921.5	28	104.3		

Source	(seq)SS	df	MS	V-ratio(F)	Prob> F
water	841.7	1	841.7	10.93	0.0027
Residual	2079.8	27	77.0		
Total	2921.5	28	104.3		

The two one-way ANOVA corresponding to the linear models

$$Y_{ijk} = \mu + \alpha_i + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

$$Y_{ijk} = \mu + \beta_j + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

respectively, with the same values of i , j , and k as described in the above.

- Remarks on the example of unbalanced two-factor analysis

- For unbalanced data, commonly the SS in the analysis of variance are **sequential sums of squares**.
- In the sprouting barley example above, sequential SS_{water} and SS_{week} are of different values when the order of the factors changes, due to the unbalanced design.
- The one-way analysis of variance tables confirmed what we have learned from the two-way analysis of variance, that the variation in the values of the response variable are better explained by both factors.

RSS for nested models — Applications to sequential ANOVA

Given a set of data, consider a sequence of nested hypotheses

$$H_o \subseteq H_1 \subseteq \dots \subseteq H_h$$

corresponding to a sequence of nested linear regression models

$$M_o \subseteq M_1 \subseteq \dots \subseteq M_h$$

all use the same data. The corresponding model parameter spaces are also nested,

$$\Theta_o \subseteq \Theta_1 \subseteq \dots \subseteq \Theta_h$$

Define

$$RSS_i = \sum_j (\text{observation}_j - \text{fitted value under } H_i)$$

Because models are nested, larger models have smaller residuals,

$$RSS_o \geq RSS_1 \geq \dots \geq RSS_h$$

The sums of squares listed in ANOVA table are ordered as

$$\begin{array}{rcl}
 & RSS_o & - RSS_1 \\
 & & - RSS_2 \\
 & & \vdots \\
 & & - RSS_h \\
 & & - RSS_{h-1} \\
 & & - RSS_h \\
 + & & RSS_h \\
 \hline
 & & RSS_o
 \end{array}$$

where

$$\begin{array}{ll}
 RSS_i - RSS_{i+1} & \text{--- Variation explained by Model}_{i+1} \text{ that cannot be explained by Model}_i \\
 RSS_h & \text{--- Remaining unexplained variation.}
 \end{array}$$

Examples

Assume that we want to compare two nested models, for example, to compare the smaller, simpler model

$$M_S : Y_{ijk} = \mu + \alpha_i + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

with the larger, more complicated model

$$M_L : Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim i.i.d. N(0, \sigma^2)$$

We can do the comparison by testing the null hypothesis

$$H_0 : \text{the smaller model is as a good fit to the data as the larger model}$$

In terms of model coefficient parameters,

$$H_0 : \beta_j = 0, \gamma_{ij} = 0, \quad \forall i, j.$$

We may use the F-test statistic to test the significance of the B factor terms,

$$F = \frac{(RSS_S - RSS_L)/d}{S_L^2} \sim F_{d, n-p_\ell} \quad \text{under } H_0$$

where

$$d = (n - p_s) - (n - p_\ell) = p_\ell - p_s$$

is the difference of RSS degrees of freedom in the two models, and

$$S_L^2 = \frac{RSS_L}{n - p_\ell}$$

RSS and R^2

$$R^2 = 1 - \frac{RSS_{model}}{RSS_{null}} = \frac{SS_{model}}{SS_{total}} = \text{"fraction of variance explained by the model"}$$

where

- RSS_{model} is the RSS or SS_e of the current model.
- RSS_{null} is the RSS or SS_e of the model that the mean of all observations are equal, $Y = \mu + \varepsilon$.
- R^2 is also called **coefficient of determination**.
- $R^2 \in [0, 1]$, always.
- the more predictors/parameters a model has, the larger R^2 .
- *Adjusted- R^2* attempts to adjust or penalize the number of predictors in the model.

$$Adjusted-R^2 = 1 - \frac{RSS_{model}/df_{model}}{RSS_{null}/df_{null}}$$

- *Adjusted- R^2* $\leq R^2$, always.